Qassim University – College of Engineering

GE 213
Introduction to Engineering Design II

Semester 342
(1434, First semester)

Assignment # 1

On Chapter ??

Using MS Excel to Determine the Size and Cost of a Football Trophy

By:

Group No. ?? Team No. ??

Member	Name	Department	PIN
1	Mohammed Ahmed	Mech. Eng. Dept.	###
2			
3			

Instructor

Dr. / Prof. .....

Date: .... / ..... / 2014

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Assignment # 1

Team Assignment

Introduction to Excel Modeling

Expectations

1. Complete reading the Excel handout material (Introduction to Excel Part-I), then make a list of Excel tools you found to be useful (at least ten tools).
2. Submit a technical report to describe the overall modeling solution process to solve the attached External Problem.

External Problem: Size and Cost of a Football Trophy

The FIFA Football Federation is thinking of manufacturing a trophy for the upcoming World Cup. It is intended to be a simple but representative trophy. One of the proposals was a mere sphere that reminds one of the ball used in the game. It would be manufactured out of solid copper but gold plated. The diameter of the trophy should be at least 10 cm but should not exceed 30 cm.

You are required to

1. Construct a model that calculates the size, mass and cost of the trophy sought for the diameters given in the problem.
2. Draw a chart showing the mass and cost of the trophy as a function of the ball diameter.
3. What would be the most "appropriate" diameter and the corresponding cost of the trophy?
4. What was your criterion for identifying the most "appropriate" diameter?
5. Would you advise the organizing committee to have a trophy of 0.5 m diameter?

1. Introduction

Through the last four sessions of course GE 213 we have learned how to use the Excel software as an effective tool for applying engineering models. This is a report out for the first assignment in the course in which we will exercise what we have learned and try to increase our experience in performing Excel models. It is a team assignment through which a simple deterministic model is performed in an Excel workbook.

This report is divided into two main parts. The first part lists some useful tools which have been learned from Excel. The second part explains how to create an Excel model to solve an external problem. The problem to be solved in this assignment is to construct a model to calculate the size, mass, and cost of a football trophy that is spherical in shape. Based on the model, the appropriate size of the trophy is to be chosen.

The report explains the problem definition, assumptions, Heuristics, modeling steps, and model implementation. The results of the computations are provided as tables and charts. The results are discussed, whereby the questions posed in the problem are answered. The team meeting minutes and the process check are provided as appendices.

2. A list of useful tools learned from Excel

1. …
2. …
3. …

…

10.…

3. External Problem

To solve the external problem, firstly, the problem should be well defined then the assumptions are stated. The Heuristics that might be helpful for solving the problem are outlined. The mathematical equations employed in the calculations are then provided. Finally, the Excel model will be implemented.

Upon completing the excel model, different sections of the implemented model will be presented as well as an overall view of the Excel model. In addition, a figure showing the formula sheet used in excel is given.

The results of the calculations are demonstrated in tables and charts. Finally, the questions posed in the external problem regarding the trophy are discussed.

3-1 Problem Definition

1. It is required to design a trophy.
2. The trophy is a sphere of diameter D.
3. It is coated with gold for only some microns.
4. A cross-sectional view of the trophy can be represented as in Fig. 1.

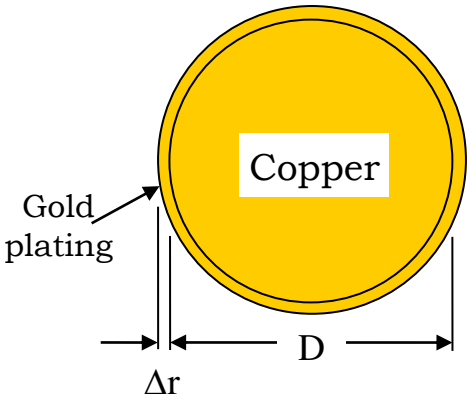


Fig. 1 Cross-sectional view of

5. **Given Data:**
   a) Minimum diameter of the Trophy = 10 cm.
   b) Maximum diameter of the Trophy = 30 cm.

3-2 Assumptions

1. The thickness of gold plating is negligible, so it does not affect neither the size nor the mass of the trophy.
2. The cost of gold plating will be included in calculating the cost of the trophy.
3. The optimum trophy size and cost is to be based on its mass as discussed in the next point.
4. The most appropriate mass of the trophy is to be less than or equal to 10 kg, since the player would not feel comfortable holding a trophy heavier than 10 kg.

3-3 Heuristics

Here are some heuristics might be helpful for solving the problem:

1. Rephrasing the problem to make sure it has been understood clearly.
2. Drawing simple diagrams to describe the problem.
3. Formulating the problem mathematically, which include:
   a) Calculating the volume and surface area of the trophy.
   b) Calculating the mass of the trophy.
   c) Calculating the cost of both copper and gold plating.

d) Calculating the total cost of the trophy.

4. Find necessary information to solve the problem: the price of each Kilo gram of copper, the density of the copper, and the price of each square meter of gold plating.

3-4 Modeling Steps and Mathematical Equations

The calculations that will be done to get answers to the questions of the problem include finding:

1. The volume of the sphere (trophy).
2. The surface area of the sphere.
3. The mass of the trophy.
4. The price of the copper sphere.
5. The cost of gold plating.
6. The total cost of the trophy.

The equations that will be used in the calculations are obtained from handbooks [2]. These include:

1) Radius of a circle:

$r = D / 2$

Where:

r: radius of the trophy, m
D: diameter of the trophy, m

2) Volume of a sphere:

$V = (4/3)\ \pi r^3$

Where:

V: volume of the trophy, m^3
π: pi (=3.14)
r: radius of the trophy, m

3) Surface area of a sphere:

$A = 4\ \pi r^2$

Where:

A: area of the trophy's surface, m^2, which is approximately the surface area of the copper ball since the thickness of coating by gold plating is negligible.

r: radius of the trophy, m

4) Mass of a sphere:

$$m = \rho V$$

Where:

m: mass of the trophy, kg
ρ: density of the trophy, kg/m^3, which is practically the density of copper because the mass of gold plating is negligible.
V: *Volume of* the trophy, m^3

5) Cost of the copper ball:

$$c_{co} = m\, p_{co}$$

Where:

c_{co}: Cost of the copper ball, SR
m: mass of the trophy, kg
p_{co}: price of each Kg of copper, SR/kg

6) Cost of the gold plating:

$$c_{gp} = A\, p_{gp}$$

Where:

c_{gp}: cost of gold plating for the trophy, SR
A: area of the trophy's surface, m^2
p_{gp}: price of gold plating, SR/kg

7) Total cost of the trophy:

$$c_{tot} = c_{co} + c_{gp}$$

Where:

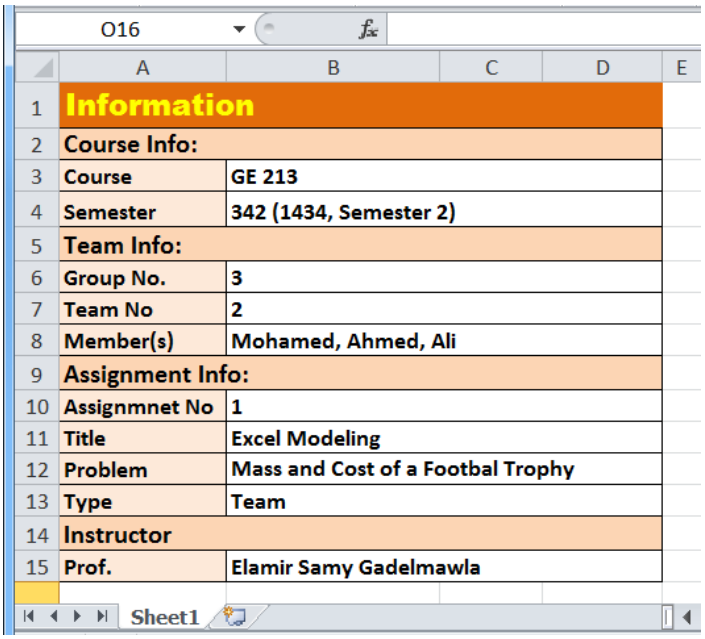
c_{tot}: total cost of the trophy, SR
c_{co}: Cost of the copper ball, SR
c_{gp}: cost of gold plating for the trophy, SR

3-5 Model Implementation

To implement the model, an Excel sheet will be created including the main four sections: Information area, Date area, Calculation area, and the charts. The Information area and Date area are presented in this section, however, the calculation area and the charts are discussed in the section 3-6 (Results and Discussions).

3-5-1 Information area

The information area of the created model is shown in Fig. 2. It includes information about the course, the team, the assignment, and the instructor.



	A	B	C	D	E
1	**Information**				
2	Course Info:				
3	Course	GE 213			
4	Semester	342 (1434, Semester 2)			
5	Team Info:				
6	Group No.	3			
7	Team No	2			
8	Member(s)	Mohamed, Ahmed, Ali			
9	Assignment Info:				
10	Assignmnet No	1			
11	Title	Excel Modeling			
12	Problem	Mass and Cost of a Footbal Trophy			
13	Type	Team			
14	Instructor				
15	Prof.	Elamir Samy Gadelmawla			

Fig. 2 Information area of the Excel model

3-5-2 Data area

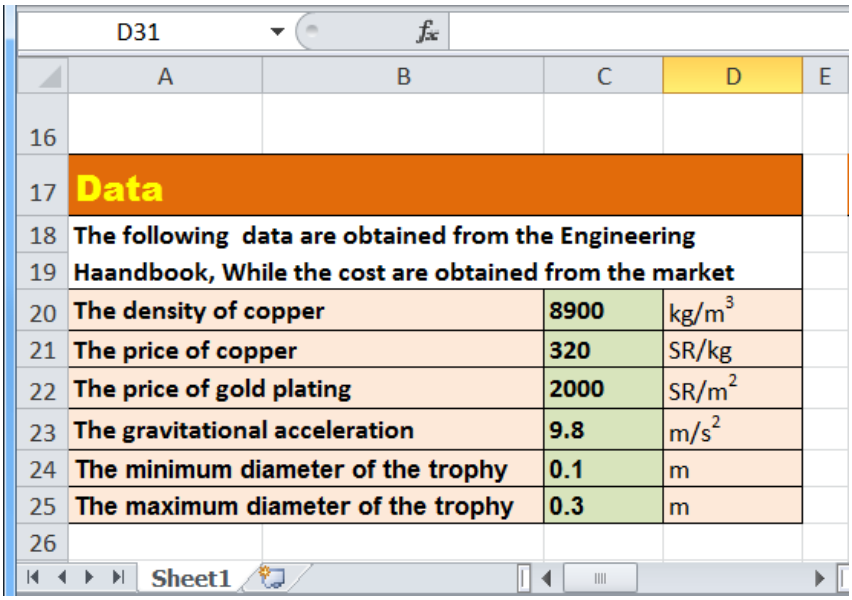
The values that will be used in the equations are obtained from handbooks [3], while the price of copper is obtained from the internet [4]. The price of gold plating is obtained from private conversations [5]. The data obtained is as follows:

The density of copper = 8,900 kg/m^3

The price of copper = 320 SR/kg

The price of gold plating = 2,000 SR/m^2

This data can be entered to the Excel model through the *Data Area* as shown in Fig 3.

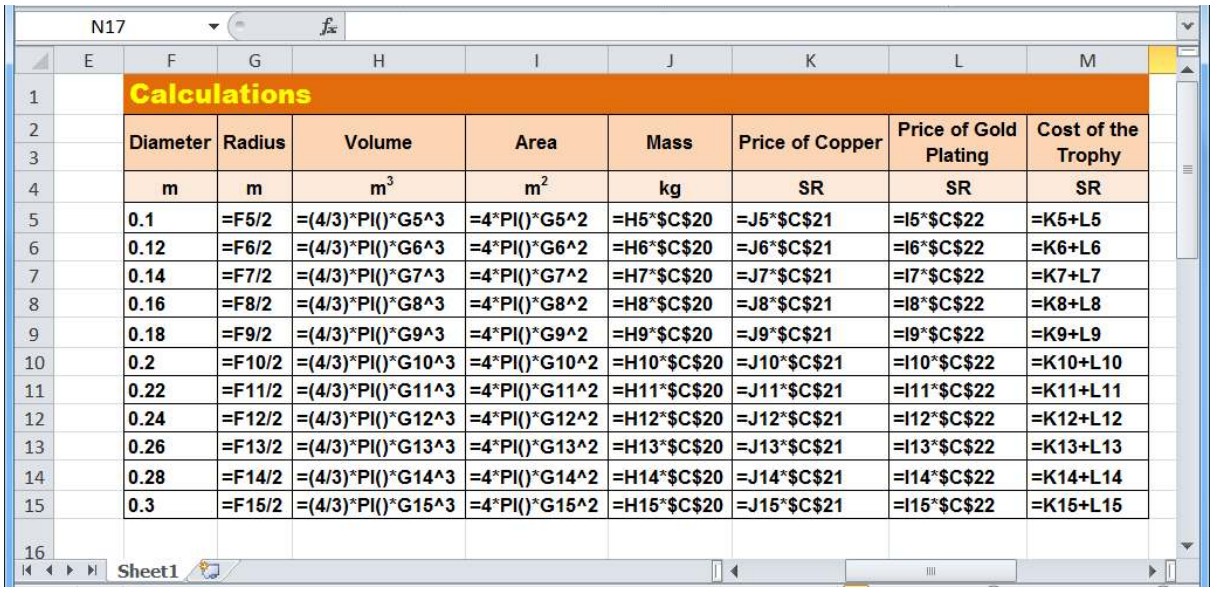


	A	B	C	D
16				
17	**Data**			
18	**The following data are obtained from the Engineering**			
19	**Haandbook, While the cost are obtained from the market**			
20	**The density of copper**		**8900**	kg/m^3
21	**The price of copper**		**320**	SR/kg
22	**The price of gold plating**		**2000**	SR/m^2
23	**The gravitational acceleration**		**9.8**	m/s^2
24	**The minimum diameter of the trophy**		**0.1**	m
25	**The maximum diameter of the trophy**		**0.3**	m
26				

Fig. 3 Data area of the Excel model

3-5-3 Excel Model Formulas

The obtained mathematical equations were entered to Excel to create a model to calculate the mass and volume of the trophy. Fig. 4 shows the formulas entered to Excel.



	E	F	G	H	I	J	K	L	M
1		**Calculations**							
2		**Diameter**	**Radius**	**Volume**	**Area**	**Mass**	**Price of Copper**	**Price of Gold Plating**	**Cost of the Trophy**
3									
4		**m**	**m**	**m^3**	**m^2**	**kg**	**SR**	**SR**	**SR**
5		0.1	=F5/2	=(4/3)*PI()*G5^3	=4*PI()*G5^2	=H5*C20	=J5*C21	=I5*C22	=K5+L5
6		0.12	=F6/2	=(4/3)*PI()*G6^3	=4*PI()*G6^2	=H6*C20	=J6*C21	=I6*C22	=K6+L6
7		0.14	=F7/2	=(4/3)*PI()*G7^3	=4*PI()*G7^2	=H7*C20	=J7*C21	=I7*C22	=K7+L7
8		0.16	=F8/2	=(4/3)*PI()*G8^3	=4*PI()*G8^2	=H8*C20	=J8*C21	=I8*C22	=K8+L8
9		0.18	=F9/2	=(4/3)*PI()*G9^3	=4*PI()*G9^2	=H9*C20	=J9*C21	=I9*C22	=K9+L9
10		0.2	=F10/2	=(4/3)*PI()*G10^3	=4*PI()*G10^2	=H10*C20	=J10*C21	=I10*C22	=K10+L10
11		0.22	=F11/2	=(4/3)*PI()*G11^3	=4*PI()*G11^2	=H11*C20	=J11*C21	=I11*C22	=K11+L11
12		0.24	=F12/2	=(4/3)*PI()*G12^3	=4*PI()*G12^2	=H12*C20	=J12*C21	=I12*C22	=K12+L12
13		0.26	=F13/2	=(4/3)*PI()*G13^3	=4*PI()*G13^2	=H13*C20	=J13*C21	=I13*C22	=K13+L13
14		0.28	=F14/2	=(4/3)*PI()*G14^3	=4*PI()*G14^2	=H14*C20	=J14*C21	=I14*C22	=K14+L14
15		0.3	=F15/2	=(4/3)*PI()*G15^3	=4*PI()*G15^2	=H15*C20	=J15*C21	=I15*C22	=K15+L15
16									

Fig. 4 Equation used in the Excel Model

3-5-4 Overall View of the Excel Model

The overall view of the excel model created to solve the external problem is shown in Fig. 5.

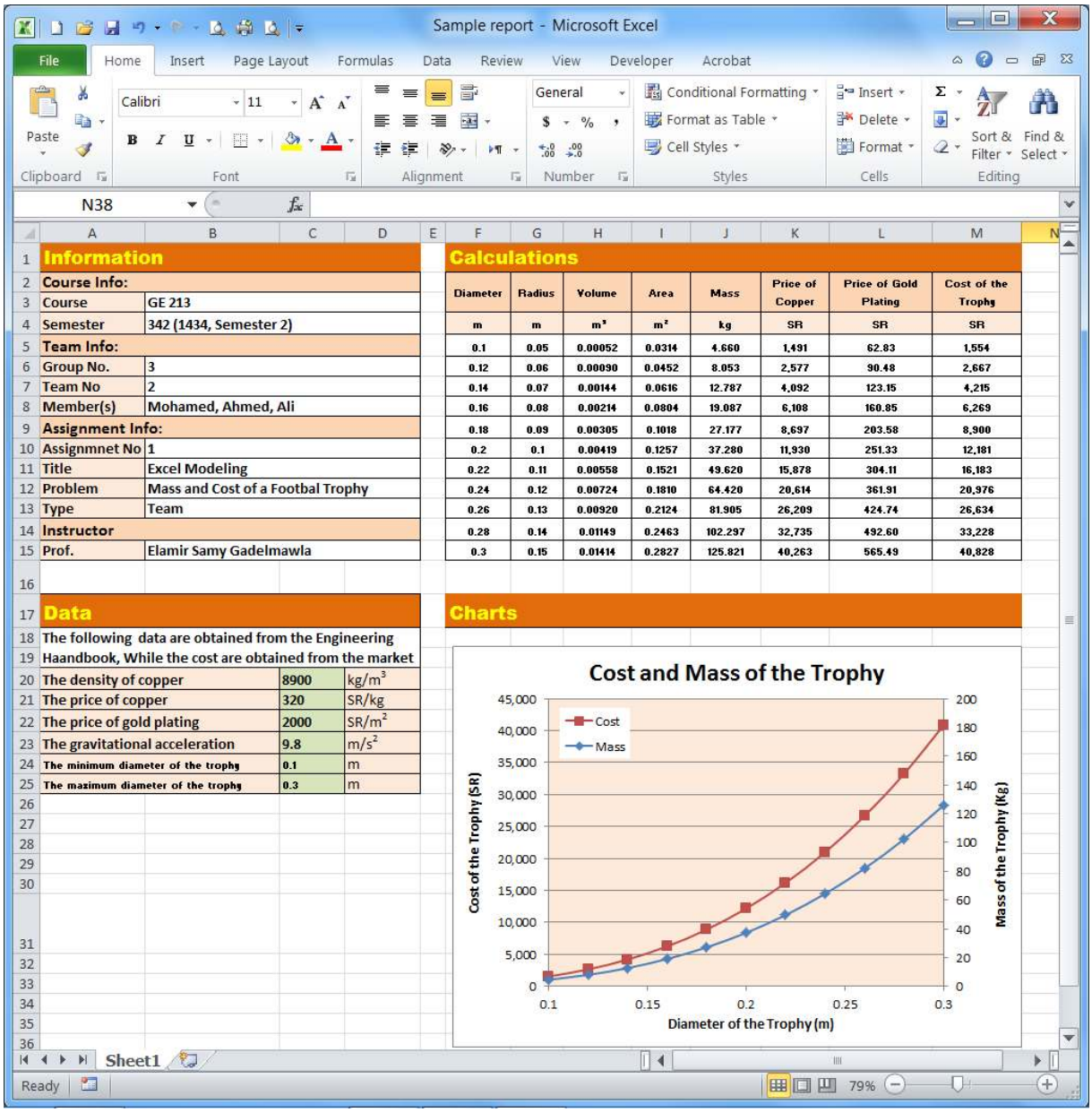


Fig. 5 The overall excel model for solving the external problem

3-6 Results and Discussion

This section explains the results obtained from the constructed model and the answers to the external problem questions.

3-6-1 Answer of Question #1

The results obtained from the constructed model include values of the diameter, radius, volume, surface area and mass. The prices of the copper ball and its gold plating are then calculated. These results are shown in Table 1.

Table 1 Results for calculating the mass and cost of the trophy

Calculations							
Diameter	Radius	Volume	Area	Mass	Price of Copper	Price of Gold Plating	Cost of the Trophy
m	m	m^3	m^2	kg	SR	SR	SR
0.1	0.05	0.00052	0.0314	4.660	1,491	62.83	1,554
0.12	0.06	0.00090	0.0452	8.053	2,577	90.48	2,667
0.14	0.07	0.00144	0.0616	12.787	4,092	123.15	4,215
0.16	0.08	0.00214	0.0804	19.087	6,108	160.85	6,269
0.18	0.09	0.00305	0.1018	27.177	8,697	203.58	8,900
0.2	0.1	0.00419	0.1257	37.280	11,930	251.33	12,181
0.22	0.11	0.00558	0.1521	49.620	15,878	304.11	16,183
0.24	0.12	0.00724	0.1810	64.420	20,614	361.91	20,976
0.26	0.13	0.00920	0.2124	81.905	26,209	424.74	26,634
0.28	0.14	0.01149	0.2463	102.297	32,735	492.60	33,228
0.3	0.15	0.01414	0.2827	125.821	40,263	565.49	40,828

3-6-2 Answer of Question #2

Fig. 6 shows a chart represents the mass and cost of the trophy as a function of the ball diameter.

3-6-3 Answer of Question #3

To calculate the most appropriate diameter of the trophy, it is assumed that the trophy should be hand carried by the team members; therefore, its mass should not exceed 10 kg. From the results presented in Table 1, the most appropriate diameter of the trophy is 12 cm as it corresponds to the mass closest to and less than 10 kg.

Therefore:

The mass of the trophy is 8.05 kg

The total price is 2,666 SR.

However, one could use a finer step size of the diameter in the calculations. This would result in a little bit higher mass of the trophy (corresponding to a larger volume due to the bigger diameter). The price of the trophy should get more expensive accordingly.

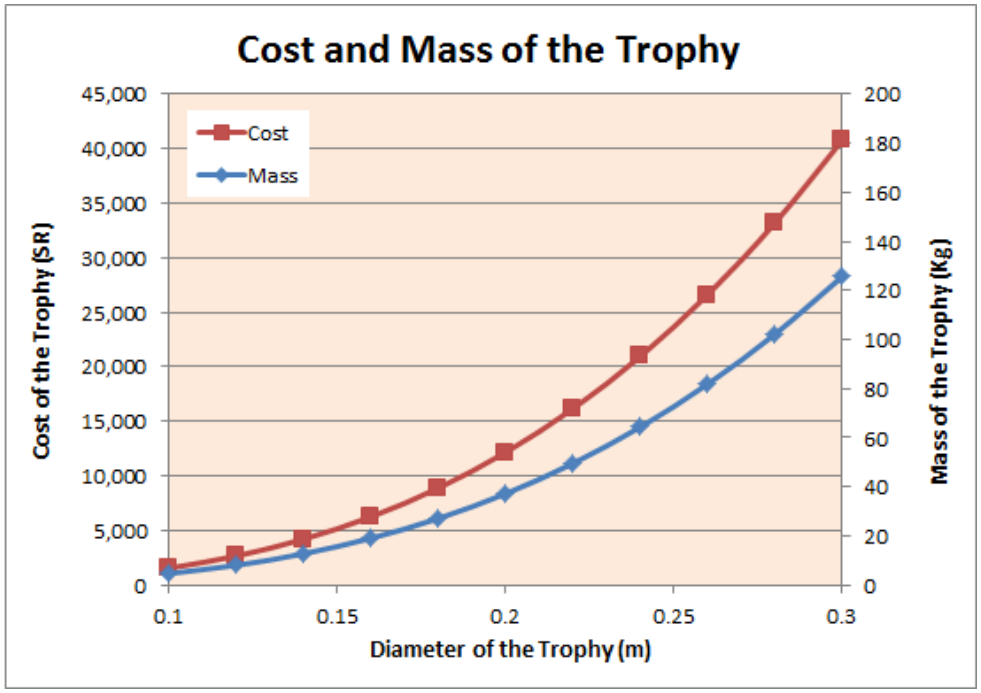


Fig. 6 calculated mass and cost of the trophy according to its diameter

3-6-4 Answer of Question #4

The criterion for identifying the most appropriate diameter was that the trophy should be hand carried by the team members.

3-6-5 Answer of Question #5

For a trophy of 0.5 m diameter:

The mass will be **582.504** kg (too heavy)

And the total price will be **187,972** SR (too expensive).

Therefore, **I do not advise** the organizing committee to have a trophy of 0.5 m diameter.

4. Conclusions

A list of the useful tools in Excel has been listed. In addition, a deterministic model had been created to calculate the size, mass and cost of a football trophy. The trophy was basically a gold-plated sphere that is made out of copper. Based on a mass not exceeding 10 kg, the values of the appropriate diameter and cost of the trophy were found to be 0.12 m and 2,666 SR, respectively. In addition, for a trophy of 0.5 m diameter, the calculated mass is 582.504 kg and the total price is 187,972 SR, therefore, I do not advise the organizing committee to have a trophy this diameter.

5. References

1. First day materials of the GE 213 course.
2. http://www.qec.edu.sa/eng/students/lectures/attachments/51/GE213/Assignment-1%20Excel%20Modeling%20-%20Term%20342.pdf, visited on 12/05/2013, 10:35 PM.
3. Korn, G. and Korn, T., Mathematical Handbook for Scientists and Engineers, 2nd Ed., McGraw-Hill, 1968.
4. Baumeister, E., Avallone, E., and Baumeister, T III, Marks' Standard Handbook for Mechanical Engineers, 8th Ed., McGraw-Hill, 1978.

Appendix I

Team Meeting Minutes

MEETING NO. 1

Time: 9:30 a.m. **Date:** 24-11-2008 (Wednesday)

Place: Fahd's house

Agenda of the meeting

duration	Subject	Responsible
9:30 – 9:40	Contact before meeting	--
9:40 – 10:00	First subject ????	??

The meeting discussions and minutes

- **First subject ????**
  Duration (??)
  The Important notes
- **Second subject ????**
  Duration (??)
  The Important notes

Decisions and recommendations of the meeting

No.	Decisions and recommendations	Responsible	Timing	Notes
1	??????	??		
2		??		

Minutes are written by : **Fahd**

Reviewed and agreed by: All members of Team (1)

Abdullah **Mohammed**